

Subin Choi

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Education

Hongik University, MS in Mechanical Engineering

Mar 2024 – Aug 2026

- GPA: 3.92/4.0

Hongik University, BS in Mechanical Engineering

Mar 2018 – Feb 2024

- GPA: total 3.65/4.0 (upper division 3.75/4.0)
- **Coursework:** Mechanical Computer Aided Engineering, Applied Computational Mechanics, Heat Transfer, Thermal Dynamics, Fluid Dynamics

Publications

1. **S. Choi**, C. Jung, D. H. Lee, and J. Choi, “Chemical Kinetic Analysis of the Pyrolysis Process of Waste Plastic-Derived Oil in a Hydrogen Plasma Reactor,” *Fuel*. **421** 139022 (2026). doi:10.1016/j.fuel.2026.139022
2. **S. Choi**, C. Jung, D. H. Lee, J. Choi, and J. Kim, “Numerical Modeling of n-Hexane Pyrolysis with an Optimized Kinetic Mechanism in a Hydrogen Plasma Reactor,” *Phys. Scr.* **101** 055007 (2026). doi:10.1088/1402-4896/ae3c67
3. E. I. Fitriasari, H. Kim, D. H. Lee, H. Song, M. Jeong, H. Lee, J. Choi, Y.-N. Kim, C. Jung, D. Cho, **S. Choi**, T. A. Atsbha, J. J. Liu, and C.-J. Lee, “Sustainability Assessment of Electrified Naphtha Cracking Pathways
4. **S. Choi**, H.-J. Park, and J. Kim, “A Review of Non-Newtonian Flows Around a Sphere,” *Korea-Aust. Rheol. J.* **37**(4), 389–403 (2025). doi:10.1007/s13367-025-00134-4
5. H. S. Choi, S. Choi, **S. Choi**, D. H. Lee, H.-D. Kim, M.-S. Kim, D.-K. Kang, and J. Choi, “Rapid Start-Up Ammonia Cracking for On-demand Hydrogen Using Rotating Gliding Arc Plasma–Catalyst Coupling,” *Int. J. Hydrogen Energy* (under review).
6. J. Choi, **S. Choi**, D. H. Lee, and I. Park, “Characterization and Mitigation of Electrode-Derived Nanoparticles in a Rotating Gliding Arc Reactor,” *Plasma Sources Sci. Technol.* (under review).
7. H. S. Choi, S. Choi, **S. Choi**, S. U. Hassan, G. Kim, D. H. Lee, M. S. Bak, and J. Choi, “Nitrogen Fixation Using a Pin-to-Pin Arc Plasma: Effects of Operating Parameters on NOx Formation Characteristics and Arc Stability,” *Chem. Eng. J.* (under review).

Research Project

Plasma-Based Nitrogen Fixation Using Pin-to-Pin Arc Reactor, Research Assistant

Nov 2025 – Present

- Investigated NOx formation characteristics in a pin-to-pin arc plasma reactor under atmospheric pressure.
- Analyzed the effects of key operating parameters—including arc gap, swirl intensity, gas flow rate, oxygen concentration, and delivered power—on NO formation, selectivity, and energy cost.
- Characterized arc dynamics using high-speed imaging combined with spectral proper orthogonal decomposition (SPOD).
- Evaluated plasma thermal state and reactive species distribution using optical emission spectroscopy (OES) and sCMOS imaging.
- Submitted the research outcomes to *Chemical Engineering Journal* (under review).

Development of Novel Plasma Process for Hydrocarbon Cracking, Research Assistant

Mar 2024 – Aug 2026

- Built a numerical framework for hydrogen plasma-assisted n-hexane pyrolysis using an optimized reaction

mechanism for high-temperature chemical kinetic analysis.

- Quantified the influence of operating parameters, including temperature, residence time, and feed composition, on conversion behavior and product selectivity.
- Analyzed reaction pathways and hydrocarbon product distributions to identify dominant decomposition characteristics in plasma-assisted cracking.
- Presented the work at international conferences based on the research findings.
- Published the main research outcomes in *Physica Scripta*.
- Contributed to a related publication in *Chemical Engineering Journal*.

Development and Integration of Platform for Organic Waste Conversion Processes, Research Assistant

Mar 2024 – Jun 2025

- Developed a chemical-kinetic analysis framework for waste plastic-derived oil pyrolysis in a hydrogen plasma reactor, focusing on reaction pathways, product distribution, and operating-condition effects.
- Performed numerical modeling and interpretation of pyrolysis experiments to explain decomposition characteristics and hydrocarbon product formation.
- Presented the study at international conferences, where the work received a **Best Oral Presentation Award**.
- Contributed to a peer-reviewed publication in *Fuel*.

Design of Computer Air Vent Using Flow Analysis Inside the Computer Body, Team Member

Mar 2023 – Dec 2023

- Performed CFD simulations to analyze airflow and thermal distribution within a computer case.
- Optimized the placement of inlets and outlets to enhance cooling efficiency.
- Predicted up to 400% improvement in heat dissipation compared to baseline design.

Future Autonomous Vehicle Driving Competition, Team Member

Dec 2022 – Feb 2023

- Implemented autonomous driving algorithms using ROS, LiDAR, and camera-based perception.
- Developed obstacle detection, lane recognition, and path planning modules.
- Optimized control algorithms for stable navigation in real test environments.
- Achieved 4th place in the national competition.

Development of an Eco-Friendly Solar Heater System Using a Fresnel Lens and Paraffin, Team Member

Mar 2022 – Dec 2023

- Designed and built a solar heating prototype using Fresnel lens concentration and paraffin-based PCM (Phase Change Material) for heat storage.
- Conducted thermal analysis and experimental validation of heat storage/discharge efficiency.
- Modeled system performance using CAD and ANSYS tools to optimize thermal behavior.
- Project recognized in university Capstone Design Competition with award placement.

Experience

Research Assistant, Korea Institute of Machinery and Materials

Mar 2024 – Feb 2026

- Investigated plasma-assisted hydrocarbon cracking for light olefin production.
- Conducted CFD simulations and kinetic modeling to analyze thermal-fluid and chemical behavior in high temperature plasma pyrolysis reactors.
- Validated simulation models with experimental data to improve accuracy.
- Drafted manuscripts and prepared presentations for international conferences.

Undergraduate Research Assistant, Hongik University

Mar 2023 – Feb 2024

- Assisted in numerical and experimental studies of plasma pyrolysis for hydrocarbon conversion.

- Automated data preprocessing, visualization, and analysis using Python.
- Supported preparation of research manuscripts and lab seminars.

Undergraduate Research Opportunity Program (UROP), Hongik University, Team Member

Sep 2022 – Feb 2023

- Participated in a structured undergraduate research training program.
- Conducted experiments, collected data, and supported project documentation.
- Gained early exposure to research methodology and academic writing.

Conference

2026 KIChE Spring Meeting and International Symposium , Oral Presentation, <i>Oxygen-Lean Arc-Plasma Nitrogen Fixation Enabled by Spatially Separated O₂ Injection for High NO_x/O₂ Ratio</i>	Apr 2026
The 10th East Asia Joint Symposium on Plasma and Electrostatic Technologies for Environmental Application (EAPETEA-10) , Oral Presentation, <i>Chemical Kinetic Analysis of the Pyrolysis Process of Waste Plastic-Derived Oil in a Hydrogen Plasma Reactor</i>	Dec 2025
The 26th International Symposium on Plasma Chemistry (ISPC26) , Oral Presentation, <i>Numerical Modeling of n-Hexane Hydropyrolysis with an Optimized Kinetic Mechanism in a Rotating Gliding Arc Reactor</i>	Jun 2025
The Korean Society of Mechanical Engineers (KSME) Thermal Engineering Division 2025 Spring Conference , Oral Presentation, <i>Numerical Simulation of n-Hexane Pyrolysis in a Rotating Gliding Arc Reactor</i>	Apr 2025
The 13th International Symposium on Non-thermal/Thermal Plasma for Pollution Control & Sustainable Energy , Oral Presentation, <i>Numerical Study of Plasma Pyrolysis of n-Hexane in a Rotating Gliding Arc Reactor</i>	Nov 2024

Honors & Scholarships

Awarded Academic Excellence Scholarship , Hongik University	Autumn 2023
Awarded Academic Excellence Scholarship , Hongik University	Spring 2023
Peer Tutoring Scholarship , Hongik University Tutored underperforming students across 4 semesters as part of an academic support scholarship program.	2022 – 2023
Subject Tutoring Scholarship , Hongik University Selected as course tutor in Physics, Chemistry, and Fluid Mechanics.	2022 – 2023
Math Tutoring Volunteer Scholarship , Hongik University Provided volunteer peer tutoring in mathematics; recognized with a service-based scholarship.	Autumn 2022

Awards

Best Oral Presentation – 2026 KIChE Spring Meeting and International Symposium	2026
Best Oral Presentation – The 10th East Asia Joint Symposium on Plasma and Electrostatic Technologies for Environmental Application (EAPETEA-10)	2025

4th Place – Mechanical Engineering AI/DX Idea Competition	2024
“Numerical Solutions to Heat Transfer Problems Using Generative AI and Python.”	
3rd Place – University Comprehensive Design Competition, Hongik University	2023
3rd Place – Creative Convergence Capstone Design Competition, Hongik University	2023
4th Place – Future Automotive Autonomous Driving Competition, Future Automotive Education Council	2023
2nd Place – University Comprehensive Design Competition, Hongik University	2022
3rd Place – Creative Convergence Capstone Design Competition, Hongik University	2022

Teaching & Tutoring Experience

Peer Tutor, Hongik University	2022 – 2023
Mentored underperforming students across 4 semesters, guiding academic recovery and study strategies.	
Course Tutor, Hongik University	2022 – 2023
Assisted students in Physics, Chemistry, and Fluid Mechanics, reinforcing theoretical understanding and problem-solving.	
Volunteer Math Tutor, Hongik University	Autumn 2022
Offered one-on-one tutoring sessions in mathematics; acknowledged with a volunteer scholarship.	
Teaching Assistant – Mechanical Engineering Dept., Hongik University	Spring 2023
Supported practical sessions in CATIA software, guiding students in design modeling and CAD practice.	
Teaching Assistant – Mathematics Dept., Hongik University	2023
Graded assignments and provided feedback under faculty supervision, ensuring consistent evaluation standards.	

Additional

Technological Skills: Proficient in ANSYS FLUENT, Altair Acusolve, Catia, Creo, Inventor, Auto CAD, Python, C++ and MATLAB